Cost Audit, Numerics Coherence, and Trust Diagnostics

Fiber Raman Suppression Project

Evidence snapshot: 2026-04-26

Abstract

This note explains how the project decides whether a Raman-suppression result is numerically trustworthy. The main lesson is simple: an impressive suppression number is not enough. The scalar objective, gradient, regularizers, units, boundary behavior, and diagnostic images must all describe the same experiment. The cost-audit runs show that changing objective scale or penalty choice can change both convergence behavior and final depth. The practical outcome is a trust checklist that every future research note should inherit.

Claim Status

Status: methodology established; audit matrix incomplete
The core numerical rules are stable: report the objective surface, keep gradients on the same scale as the optimizer cost, inspect boundary and conservation checks, and include standard images. The older cost-audit matrix is useful but incomplete: one fiber/power configuration and one sharpness-aware run are still missing. This note is therefore a trustworthy-methods note, not a claim that every candidate result has been fully revalidated.

1 Question and Thesis

The research notes in this series compare many different phase masks, optimizers, basis choices, and fiber settings. Those comparisons only make sense if they are using the same meaning of "cost." This note asks:

When should a low Raman cost be trusted as evidence?

The thesis is that every reported result needs two layers of evidence:

$$\text{physics performance} \quad + \quad \text{numerical coherence.}$$

Physics performance is the Raman-band suppression itself. Numerical coherence means the objective, gradient, regularization, grid, boundary checks, and plots all agree about what was optimized.

Intuition Check

If a graph says a mask reached $-70\,\mathrm{dB}$, the first question should not be "is that good?" It should be "what exact scalar function produced that number, and did the solver really optimize that same scalar function?" This is the difference between a result and a number that happens to come out of a script.

2 The Cost Function

The physical quantity of interest is the fraction of output spectral energy in the Raman-shifted band:

$$J_{\mathrm{phys}}(\phi) = \frac{\sum_{\omega_k \in \mathcal{B}_{\mathrm{Raman}}} |u_{\omega_k}(L;\phi)|^2 \, \Delta\omega}{\sum_k |u_{\omega_k}(L;\phi)|^2 \, \Delta\omega}.$$

The shaped input is

$$u_\omega(0;\phi) = u_{\omega,0} \exp(i\phi(\omega)).$$

The field $u_\omega(L;\phi)$ comes from the nonlinear fiber propagation solver. The forward model follows the standard split-step / interaction-picture fiber-simulation literature [1, 2, 3].

The project often reports

$$J_{\mathrm{dB}}(\phi) = 10 \log_{10} J_{\mathrm{phys}}(\phi).$$

More negative values mean less Raman-band energy. A change from $-30\,\mathrm{dB}$ to $-60\,\mathrm{dB}$ is not a small cosmetic change: it is a factor of 10^3 in the Raman-band fraction.

Intuition Check

J_{phys} is a fraction. If $J_{\mathrm{phys}} = 10^{-5}$, then one part in 100,000 of the output spectral energy lies in the Raman band. The dB form is just a more readable way to display tiny fractions.

3 Linear Surface Versus Optimizer Surface

The optimizer does not always see only J_{phys}. Some runs include penalties:

$$J(\phi) = J_{\mathrm{phys}}(\phi) + \lambda_{\mathrm{gdd}} R_{\mathrm{gdd}}(\phi) + \lambda_{\mathrm{edge}} R_{\mathrm{edge}}(\phi).$$

Here R_{gdd} discourages extreme curvature in spectral phase, and R_{edge} discourages shaped pulses that put too much time-domain energy near the FFT-window boundary.

The scalar surface passed to the optimizer is either the linear surface

$$J_{\mathrm{surf}}(\phi) = J(\phi),$$

or the dB surface

$$J_{\mathrm{surf}}(\phi) = 10 \log_{10} J(\phi).$$

The important rule is that the gradient must match this exact surface:

$$\nabla J_{\mathrm{surf}} = \frac{10}{\ln 10} \frac{1}{J} \nabla J \qquad \text{when the dB surface is used.}$$

The curvature warning is the same idea one derivative later. If $F(\phi) = 10\log_{10} J(\phi)$, then the Hessian is not just the linear Hessian multiplied by a constant:

$$\nabla^2 F = \frac{10}{\ln 10}\left(\frac{1}{J}\nabla^2 J - \frac{1}{J^2}\nabla J \nabla J^T\right).$$

So a Hessian-vector product built for the linear physics cost is useful, but it is not automatically the Hessian of the regularized dB optimizer surface.

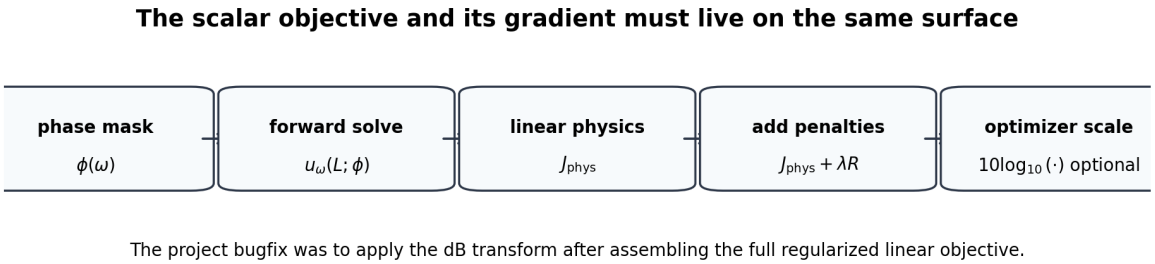


Figure 1: Objective-surface pipeline. The dB transform belongs after the full regularized linear objective has been assembled, so the cost and gradient seen by the optimizer describe the same function.

Intuition Check

The dB transform is like changing from meters to centimeters, except it is nonlinear. You cannot change the units of the cost and forget to change the gradient. If the cost is in dB but the gradient is still linear, the optimizer is following directions for the wrong map.

What The Regularizers Mean

The curvature penalty is a discrete second-difference penalty on spectral phase:

$$R_{\text{gdd}}(\phi) = \sum_{i=2}^{N_t-1} \frac{(\phi_{i+1} - 2\phi_i + \phi_{i-1})^2}{\Delta\omega^3}, \qquad \Delta\omega = \frac{2\pi}{N_t \Delta t}.$$

This is not trying to prove the phase is physically optimal. It is a numerical pressure against very sharp pixel-to-pixel curvature.

The boundary penalty measures how much shaped input energy sits near the beginning and end of the time window:

$$R_{\text{edge}}(\phi) = \frac{\sum_{t_j \in \text{edge}} |u(t_j, 0; \phi)|^2}{\sum_j |u(t_j, 0; \phi)|^2}.$$

The trust report separately checks both input and output edge fractions. This matters because a result can look deep if the pulse is being pushed into the numerical window boundary.

Intuition Check

The curvature penalty says "do not use a wildly jagged phase unless it really helps." The edge penalty says "do not win by hiding energy near the edge of the simulation box."

4 Gauge Fixing: What Phase Structure Counts?

Not every part of $\phi(\omega)$ is physically meaningful for the comparison we want to make. A constant phase does not change intensity. A linear spectral phase mainly shifts the pulse in time. Those two directions are treated as gauge-like directions:

$$\mathcal{G} = \mathrm{span}\{1, \omega - \bar{\omega}\}.$$

For shape comparisons, the project projects them out:

$$\phi_{\perp} = (I - P_{\mathcal{G}})\phi.$$

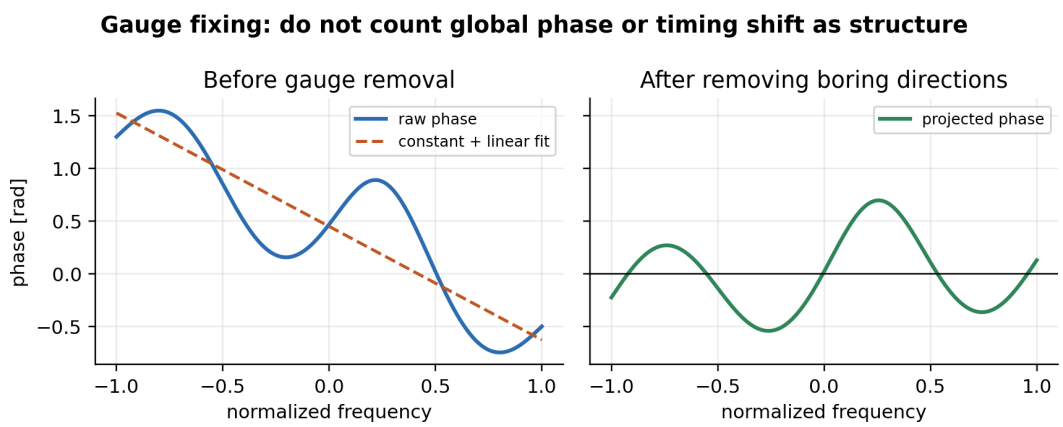


Figure 2: Gauge fixing removes the constant and linear pieces of spectral phase. The remaining curve is the part that should be compared when asking whether two masks have the same meaningful shape.

Intuition Check

Quick version: "subtract off the boring directions before comparing masks." Global phase and timing shift can make two curves look different even when they represent the same useful pulse-shaping idea.

5 Implementation Surface

Component	Numerical responsibility
Raman optimization library	Builds the single-mode Raman cost and adjoint gradient, including optional regularizers and the dB objective transform.
Objective-surface helper	Records the scalar objective surface in machine-readable form so trust reports state whether the run used a linear or dB scale.
Numerical trust reporter	Builds pass/marginal/suspect reports for determinism, boundary energy, conservation, gradient checks, and cost-surface metadata.
Standard-image helper	Writes the required phase diagnostic, optimized evolution, and unshaped control evolution images after an optimization run.
Cost-audit workflow	Runs the comparison matrix over objective variants and fiber/power settings.

The key code invariant is that regularizers are added on the linear surface first. If a dB objective is requested, the dB transform is then applied to the combined scalar objective and to the combined gradient.

6 Cost-Audit Evidence

The cost-audit matrix compared objective variants across two completed fiber/power configurations. Setup A is a short, low-power SMF-28 case ($L = 0.5\,\mathrm{m}, P = 0.05\,\mathrm{W}$). Setup B is a longer, higher-power SMF-28 case ($L = 5.0\,\mathrm{m}, P = 0.2\,\mathrm{W}$). The missing setup is an HNLF case that should not be summarized as completed yet.

The variants were: direct linear cost, dB-scaled cost, curvature-regularized cost, and a sharpness-aware cost. The point is not that one variant wins every time. The point is that changing the scalar surface changes the optimization path enough that notes must name the surface explicitly.

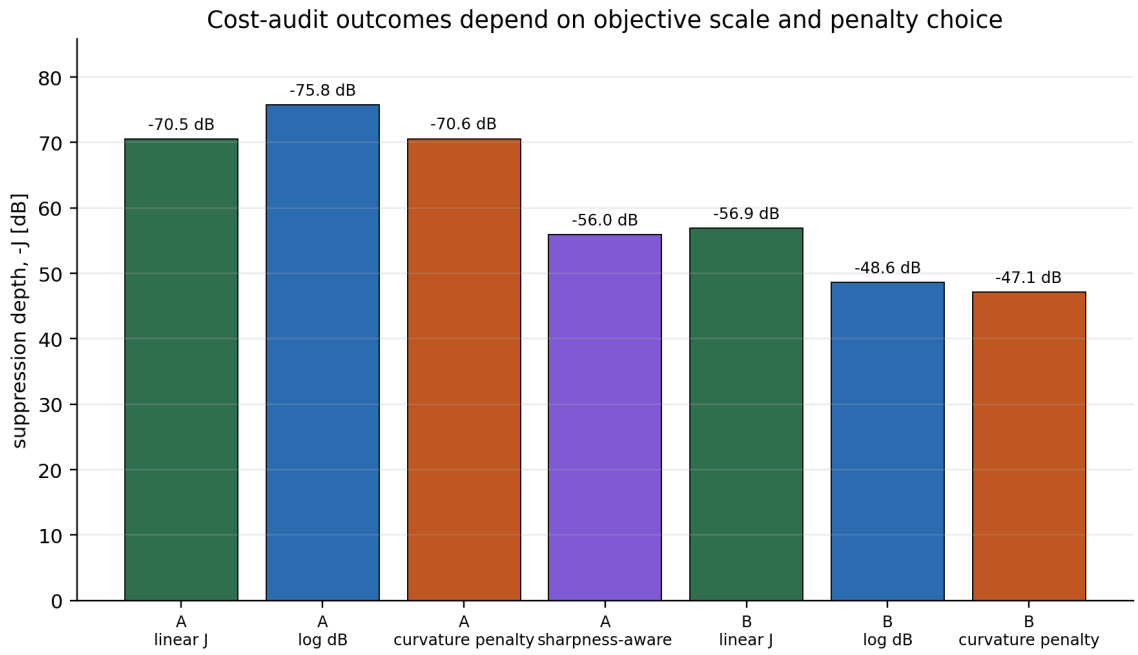


Figure 3: Completed cost-audit outcomes. The final suppression depth depends on the objective scale and penalty choice, so results should always state which surface was optimized.

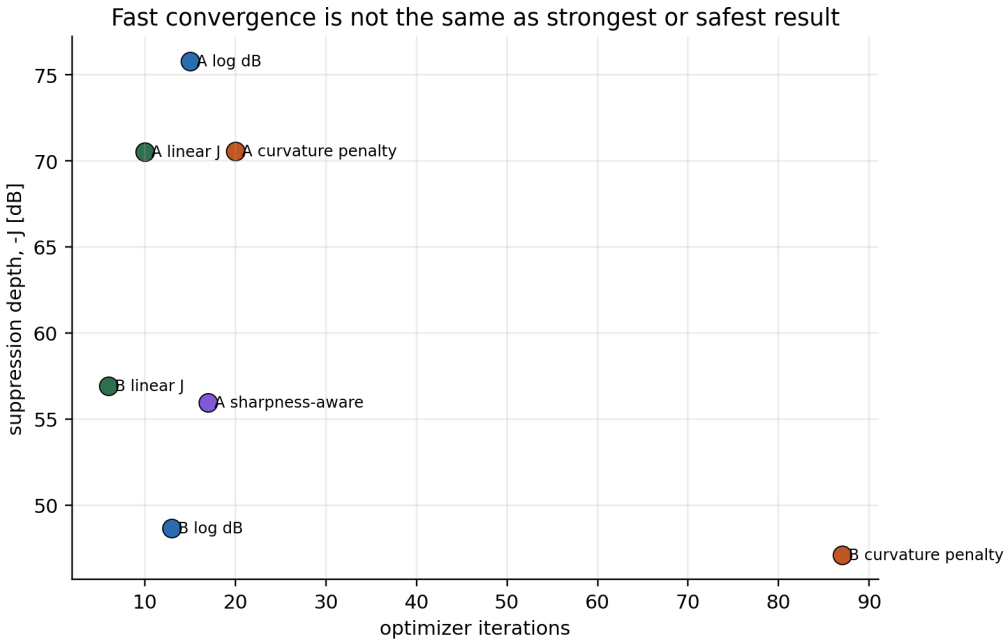


Figure 4: Iteration-depth tradeoff. Fast convergence, deepest native depth, and best numerical trust are related but not identical goals.

Setup	variant	final J	iterations	reading
A	linear J	$-70.53\,\mathrm{dB}$	10	strong and fast, but linear scale can become tiny and ill-conditioned.
A	dB surface	$-75.79\,\mathrm{dB}$	15	deepest completed result in this audit set.
A	curvature penalty	$-70.57\,\mathrm{dB}$	20	similar depth with explicit smoothness pressure.
A	sharpness-aware	$-55.96\,\mathrm{dB}$	17	shallower because the objective asks for local stability, not only depth.
B	linear J	$-56.93\,\mathrm{dB}$	6	deepest completed result for this setup.
B	dB surface	$-48.64\,\mathrm{dB}$	13	less deep here; objective scale is not automatically better in every regime.
B	curvature penalty	$-47.10\,\mathrm{dB}$	87	slowest completed run in this audit set.

Intuition Check

The audit does not say "always use the dB objective." It says "say what you used, and do not compare runs as if the objective surface were irrelevant." Changing the surface can change the path, the convergence criterion, and the final mask.

7 Trust Diagnostics

The numerical trust report turns scattered checks into a repeatable gate:

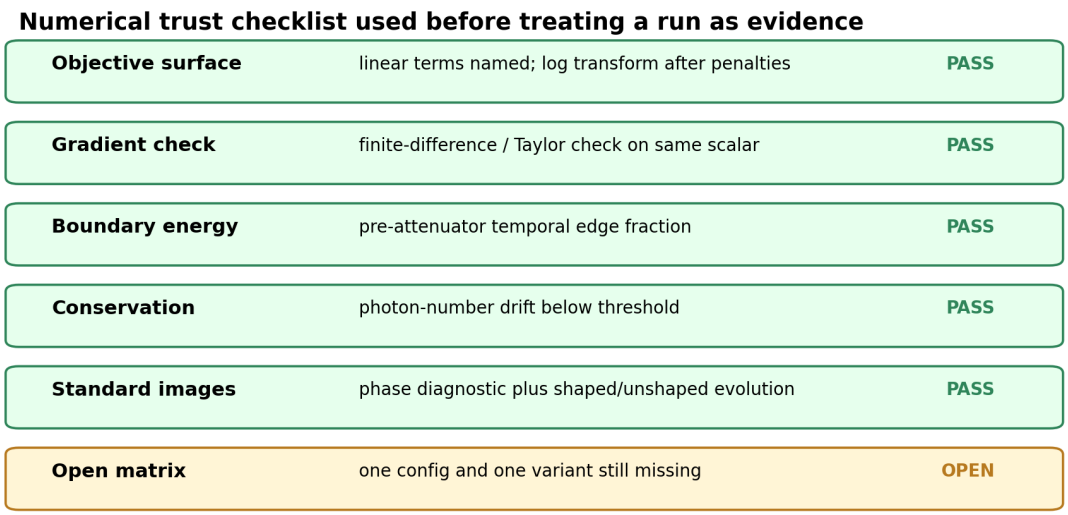


Figure 5: Trust checklist. A result can be visually exciting and still be marked open or suspect if the objective, boundary, conservation, gradient, or standard-image evidence is incomplete.

The most important checks are:

- **Objective surface:** the report must say whether the scalar surface was linear or dB-scaled, and which penalties were included.

- **Gradient validation:** finite-difference or Taylor checks must compare against the same scalar objective that the optimizer sees.
- **Boundary energy:** time-domain energy near the FFT-window edge is checked before the absorber can hide it.
- **Conservation:** photon-number drift should stay below the configured pass or marginal thresholds.
- **Standard images:** the PDF should show the phase diagnostic and corresponding spectral evolution, plus an unshaped control.

Metric	Pass	Marginal	Why it matters
Max edge fraction	$\leq 10^{-3}$	$\leq 10^{-2}$	Checks whether the pulse fits in the time window.
Photon-number drift	$\leq 10^{-4}$	$\leq 10^{-3}$	Checks whether propagation is numerically coherent.
Gradient max relative error	$\leq 5 \times 10^{-2}$	$\leq 10^{-1}$	Checks whether the optimizer direction matches the scalar cost.

Intuition Check

Think of the trust report as a lab notebook checklist. It does not prove the physics is publishable by itself, but it catches common ways a simulation can look better than it deserves.

8 Representative Standard Images

The standard images are not decoration. They are part of the audit. The phase diagnostic tells us whether the mask is smooth, wrapped, pixel-like, or dominated by gauge directions. The heat map shows whether the shaped pulse actually changes the spectral evolution relative to the control.

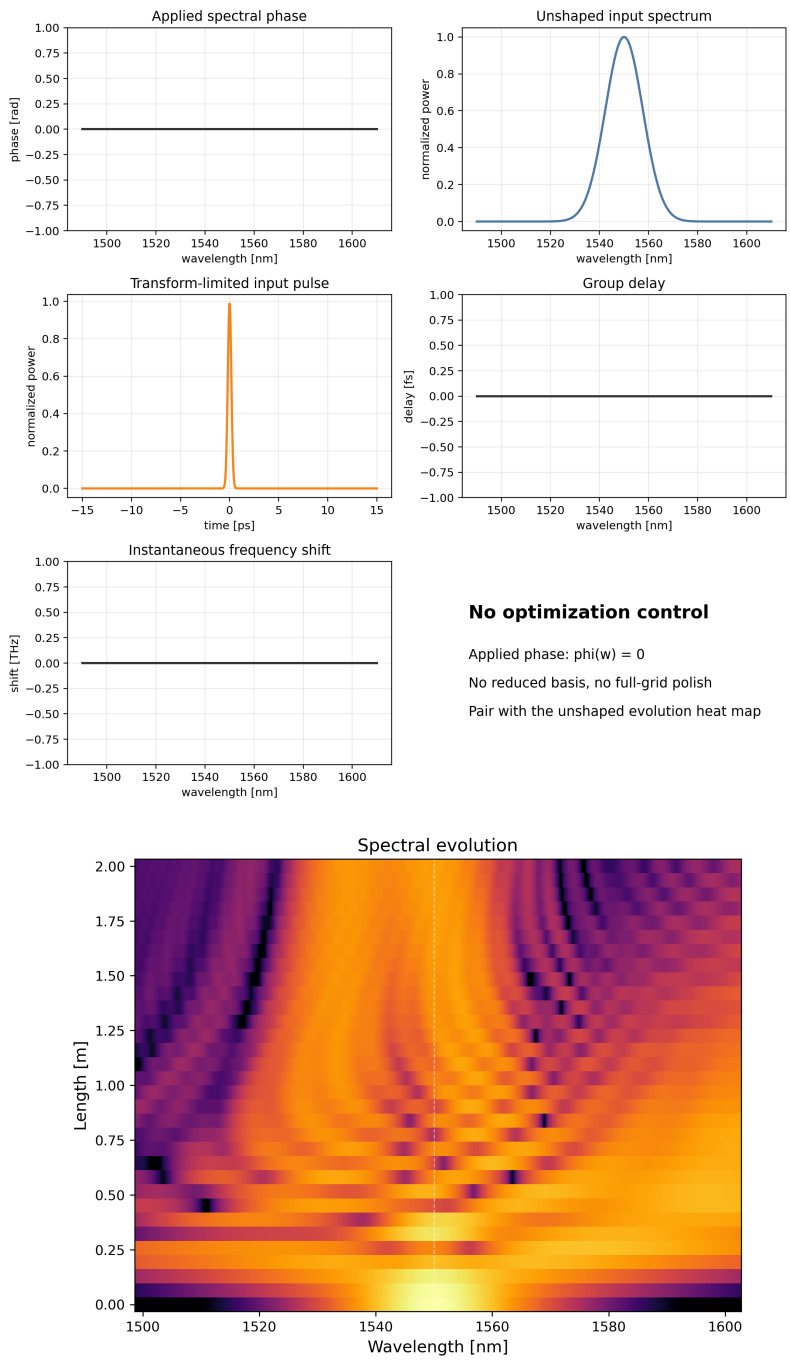
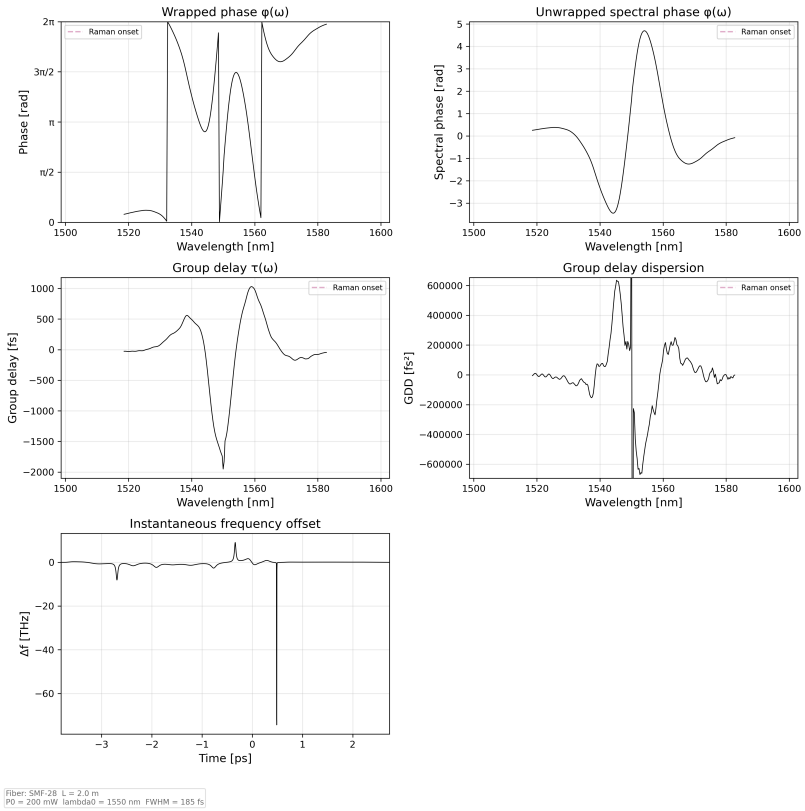


Figure 6: No-optimization control. A flat phase and the corresponding unshaped evolution give the visual reference for deciding whether a shaped result is actually suppressing Raman-band growth.



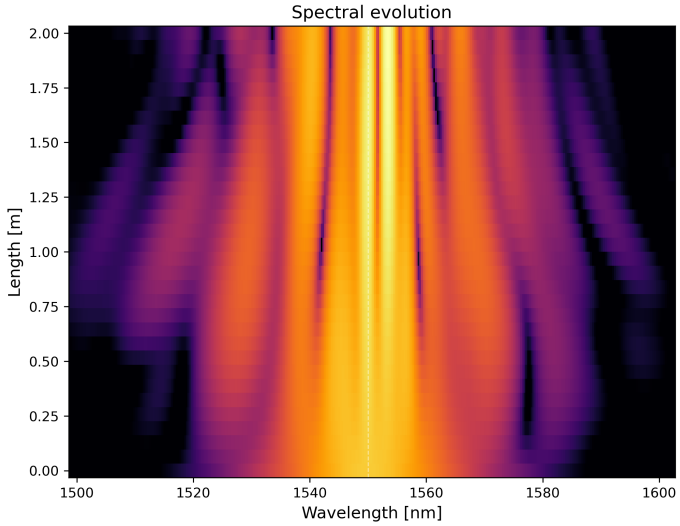


Figure 7: Representative shaped result. The phase diagnostic and heat map must be read together: the curve shows what mask was applied, and the heat map shows how the spectrum evolved under that mask.

9 Current Limits and Missing Evidence

The cost-audit lane is strong enough to define project standards, but it is not complete enough to settle every optimizer comparison.

- One planned fiber/power configuration in the audit matrix is still missing all objective variants.
- One sharpness-aware run is missing from the completed second setup.
- Hessian and trust-region tools must always state which scalar surface their Hessian-vector products represent. A linear-cost Hessian is useful, but it is not automatically the Hessian of a regularized dB objective.
- Very deep results, roughly the -60 to $-80\,\mathrm{dB}$ range, should be treated carefully because tiny linear fractions can expose solver tolerance, scaling, and conditioning issues.
- Mixed unit conventions remain a readability and conditioning hazard. The code works with them, but outward notes should be explicit about ps, THz, rad/ps, seconds, and normalized frequency whenever a formula depends on units.

Intuition Check

The right scientific posture is not pessimism. It is bookkeeping. A result is much easier to defend when the note already says what was checked, what was not checked, and what would change the conclusion.

10 Presentation Capsule

If this note becomes a methods-quality slide, the clean message is:

- The scalar objective, gradient, regularizers, and dB transform must all describe the same optimization surface.
- Gauge directions matter because constant and timing-like phase changes can look like structure without changing the physical claim.
- Standard images are a verification tool: every important phase profile should be paired with the heat map that shows what happened in propagation.
- This note is the credibility layer for the project; it is not the place to sell the most dramatic Raman before/after result.

11 Reproduction Capsule

For a future rerun, use the cost-audit driver on the heavy machine rather than running substantial simulations on the editing host:

```
julia -t auto -project=.  scripts/research/cost_audit/cost_audit_driver.jl.
```

The expected durable outputs are result payloads, a summarized audit table, trust metadata, and standard images for any run that produces an optimized phase. For documentation production, the minimum acceptable evidence package for a candidate is:

1. objective-surface statement: linear or dB, with penalties listed;
2. final Raman fraction in dB and, when needed, the underlying linear fraction;
3. gradient or Taylor validation status;
4. boundary and conservation verdicts;
5. no-optimization control image;
6. phase diagnostic plus corresponding spectral-evolution heat map.

12 Sources and References

References

[1] G. P. Agrawal. *Nonlinear Fiber Optics*, 6th ed. Academic Press, 2019. https://www.sciencedirect.com/book/9780128170427/nonlinear-fiber-optics

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